

# LEC8 - Eclipses

1/1

\*We will further investigate eclipses in Stellarium project 3

## What Causes Eclipses

- Earth & the Moon cast shadows
- When either passes through the others shadow

**Penumbra:** Partially shaded region

## Lunar Eclipses

- Lunar eclipses can be either total, partial, penumbral
- We only get lunar eclipses during full Moons
- Total lunar eclipses have penumbral & partial phases
- Total lunar eclipses can be seen from anywhere on Earth

**Umbra:** Fully shaded region

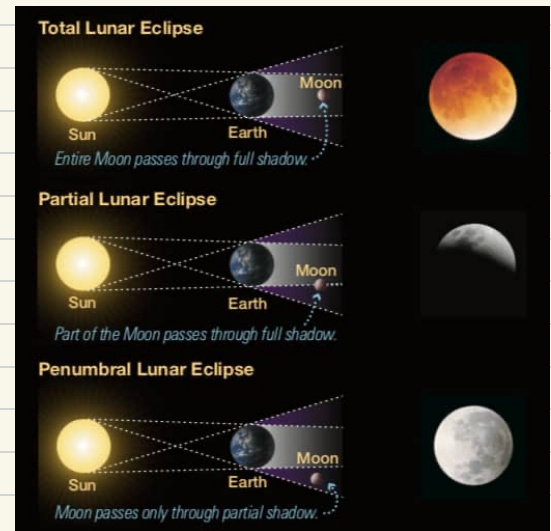
**Eccentric:** A non-circular orbit. The more eccentric an orbit is, the more oval shaped it is.

## Why does the Moon look red during a total lunar eclipse

- Sunlight passes through Earth's atmosphere, which scatters blue light  
↳ Causes mostly red light to illuminate the moon

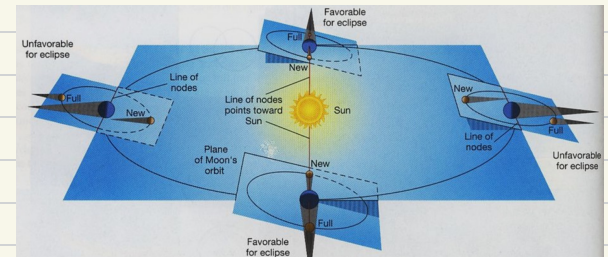
## Solar Eclipses

- Can be either total, partial, annular
- Can only occur during new Moon phase
- Total solar eclipses can only be seen in narrow region



## Why do total or annular solar eclipses occur

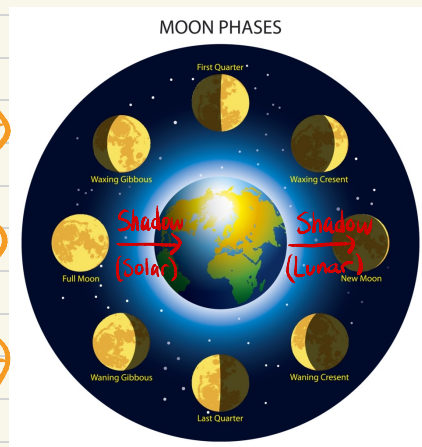
- The Moon's orbit is slightly **eccentric**
- Total solar eclipse when Moon is closer to Earth
- Annular solar eclipse when Moon is farther to Earth



## Why don't we see a solar eclipse once a month

- Eclipses occur when moon is at new **OR** full phases while **ALSO** passing through the ecliptic plane
- We can predict solar eclipses before they happen

S  
U  
N  
→  
L  
→  
I  
→  
G  
H  
T



- Solar eclipses occur when the moon casts a shadow on the Earth, and it can be total, partial, or annular.
- Solar eclipses only occur during New Moons
- Lunar eclipses occur when the Earth casts a shadow on the Moon. They can be total, partial, or penumbral
- Lunar eclipses only occur during Full Moons